Presentation Results Pack: LoRA Geometry + Oracle Routing

Experiment Overview

Experiment	Key measured result
Exp 1: Full-FT SVD (SST-2 q/v)	Mean cumulative energy at k=64 is 0.7068; k=107 reaches 80% energy.
Exp 2: LoRA-FT subspace overlap (SST-2 q/v)	Mean overlap rises from 0.0472 at k=1 to 0.1228 at k=64 (computed over 24 matrices).
Exp 3: Oracle-routed LoRA experts across 6 GLUE tasks	Weighted primary = 0.8919; vs per-task Full-FT weighted primary delta = -0.0018; end-to-end runtime = 44.61s.

Experiment 1: Full-FT Spectral Energy (SST-2)

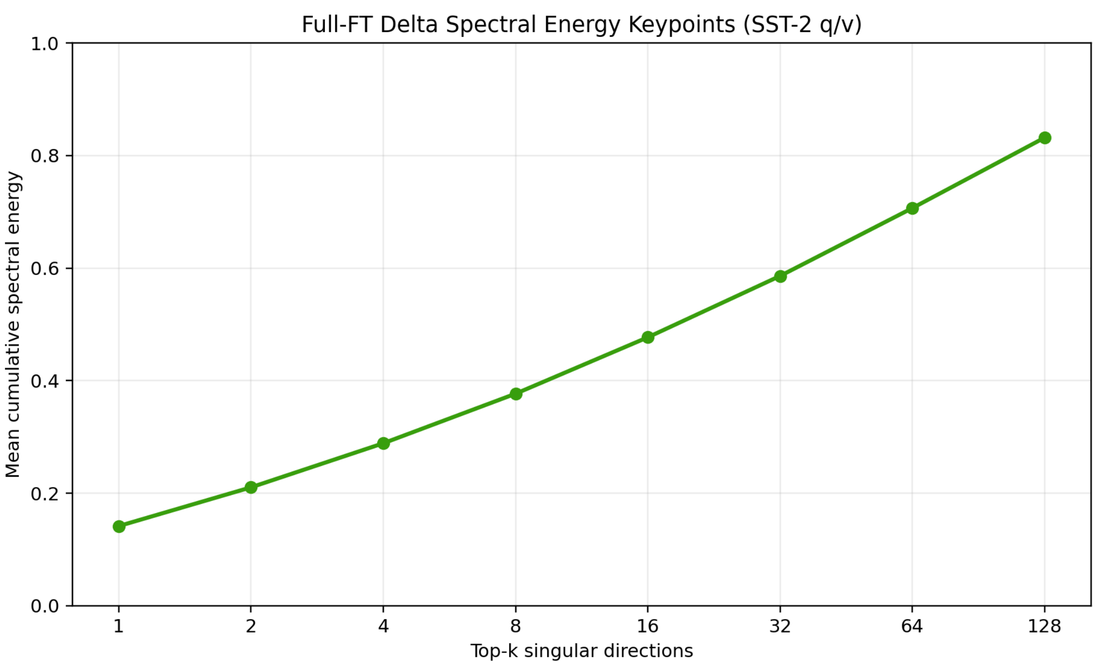

Figure 1: Mean cumulative spectral energy across SST-2 query/value update matrices.

k	Mean cumulative energy
1	0.1415
2	0.2104
4	0.2887
8	0.3768
16	0.4770
32	0.5861
64	0.7068
128	0.8320
k for 80% energy	107
k for 90% energy	128
k for 95% energy	128

Experiment 2: LoRA vs Full-FT Subspace Overlap (SST-2)

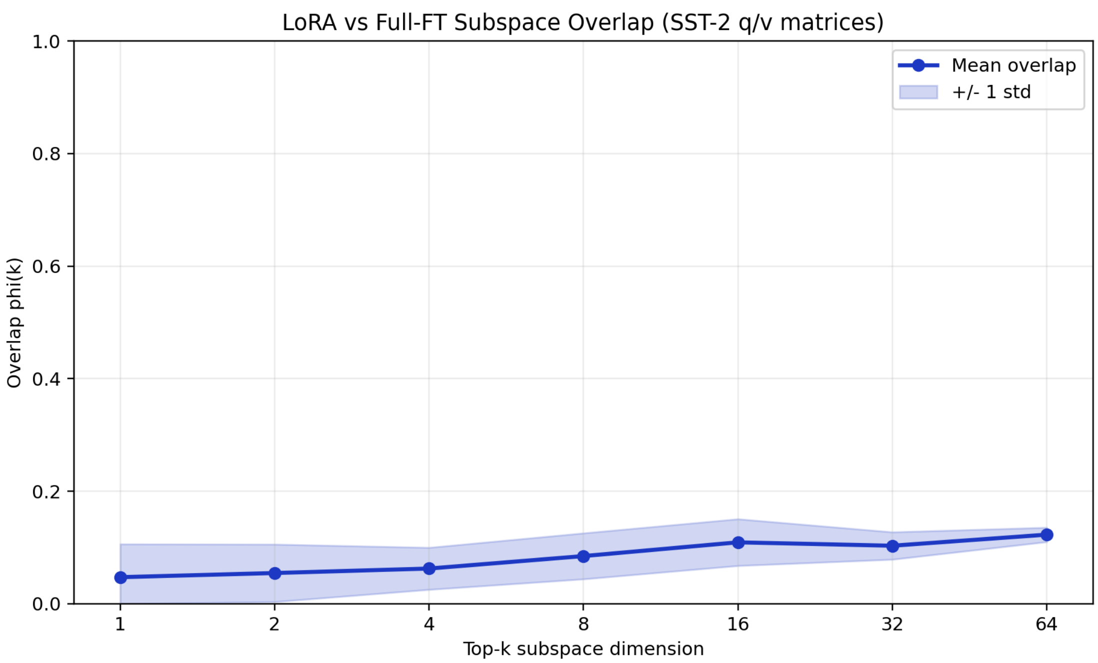

Figure 2: Subspace overlap $\phi(k)$ with mean and ±1 std across matrices.

k	Mean overlap $\phi(k)$	Std	Min	Max
1	0.0472	0.0589	0.0000	0.2592
2	0.0546	0.0510	0.0078	0.2029
4	0.0625	0.0373	0.0162	0.1636
8	0.0847	0.0408	0.0272	0.1933
16	0.1091	0.0415	0.0500	0.1837
32	0.1031	0.0243	0.0654	0.1454
64	0.1228	0.0128	0.0987	0.1433

Experiment 3: Oracle-Routed LoRA vs Baselines (6 GLUE tasks)

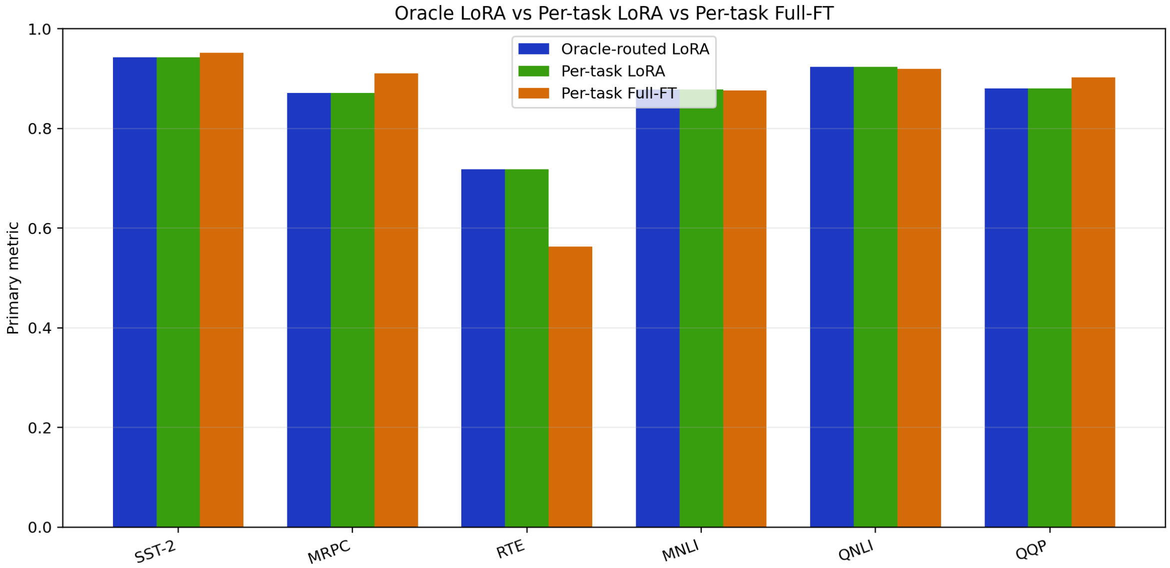

Figure 3: Primary metric comparison by task.

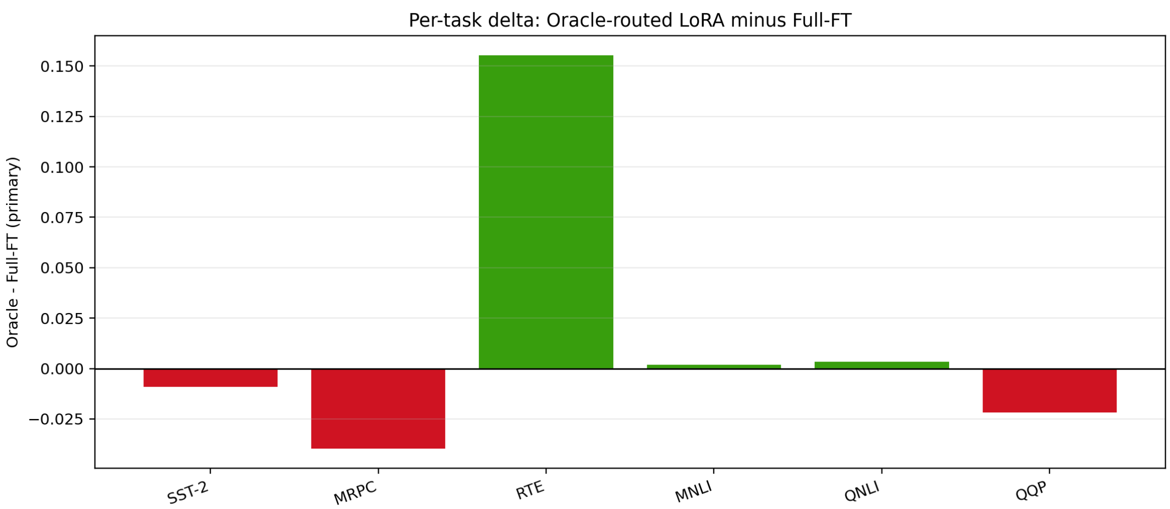

Figure 4: Per-task primary-metric delta between oracle-routed LoRA and per-task Full-FT.

Task	N	Oracle	LoRA	Full-FT	Oracle-LoRA	Oracle-FT
SST-2	872	0.9427	0.9427	0.9518	0.0000	-0.0092
MRPC	408	0.8709	0.8709	0.9105	0.0000	-0.0396
RTE	277	0.7184	0.7184	0.5632	0.0000	0.1552
MNLI	2000	0.8780	0.8780	0.8760	0.0000	0.0020
QNLI	2000	0.9235	0.9235	0.9200	0.0000	0.0035
QQP	2000	0.8804	0.8804	0.9021	0.0000	-0.0217

Metric	Oracle	LoRA	Full-FT	Oracle-LoRA	Oracle-FT
Macro primary	0.8690	0.8690	0.8539	0.0000	0.0150
Weighted primary	0.8919	0.8919	0.8937	0.0000	-0.0018
Macro accuracy	0.8690	0.8690	0.8545	0.0000	0.0145
Weighted accuracy	0.8963	0.8963	0.8974	0.0000	-0.0012